

Light Gradient Boosting Machines(LGBM)

(Lighter version of GB)

I was developed by Microsoft, April 2017.

Main Idea:

"To make Gradient Boosting on Decision Tree faster".

Points to Remember

1. Since it is a lighter version it makes more faster, more efficient and also a bit more accurate predictions.
2. It splits the **Decision Tree leaf-wise** unlike other Boosting algorithms and also Bagging algorithm using **Gini Index**.
3. LightGBM is almost **7times faster than XGBoost** and is much better approach when dealing with large datasets with a limited amount of time.
4. Since, LGBMs are growing leaf-wise, sometimes it **can lead to overfitting**. However, it can be minimized by defining the depth for splitting.
5. **Many hyperparameter** need to be optimized.
6. This is not a part of sklearn library. **Have to install separately**.
7. LightGBMs uses a technique called as **GOSS(Gradient based One Side Sampling)** for splitting the tree.
8. It is having the inbuilt **intelligence to handle the sparse dataset**.
9. It will do **Feature Bundling** to reduce the dimensions to some extent.



Light GBM
Goss

[illegible]

1

5

9

10

11

20

25

•

•

•

150

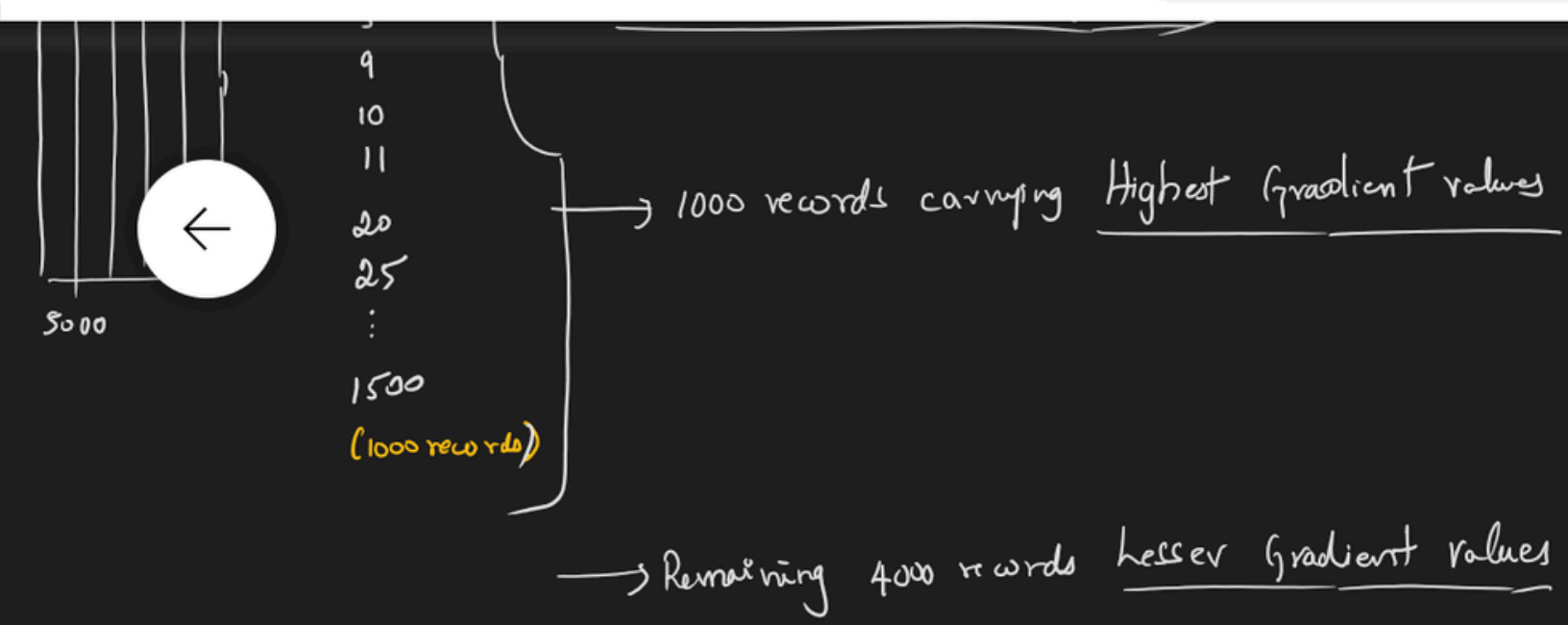
Close



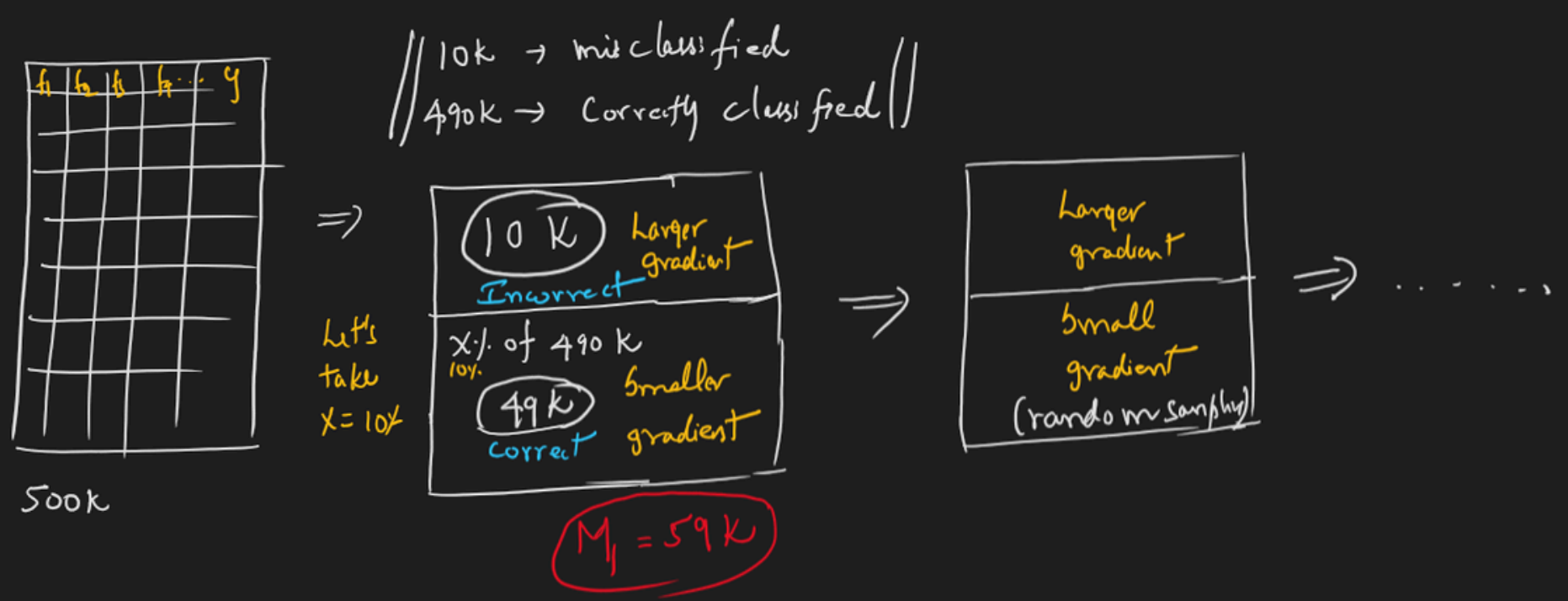
High Error Rate = High Gain

→ 1000 records carrying Highest Gradient values

→ Remaining 4000 words lesser gradient values



How exactly training is happening wrt my LG BMs?



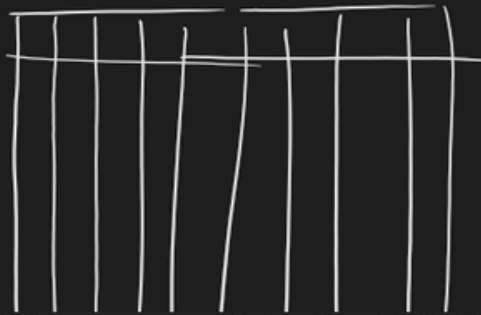
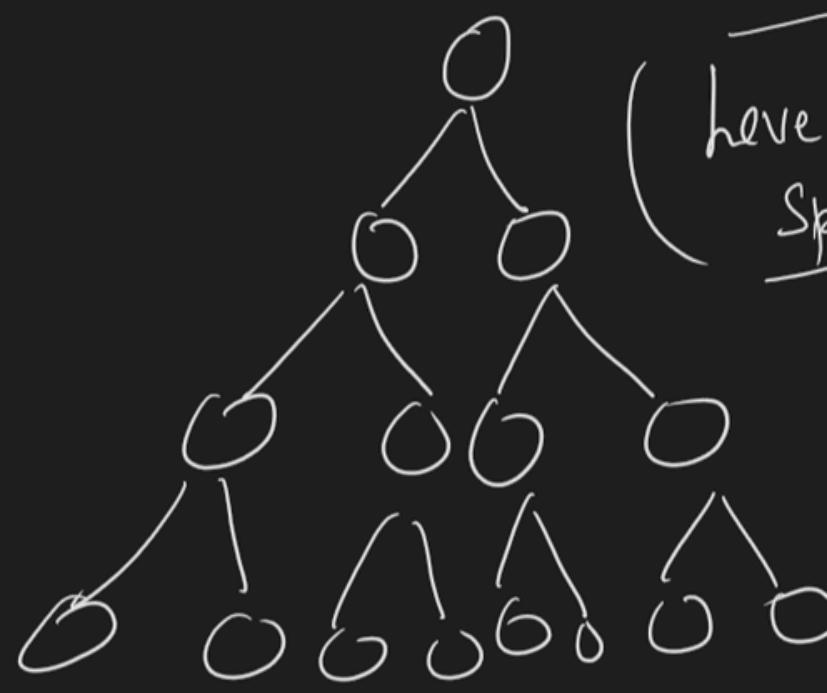
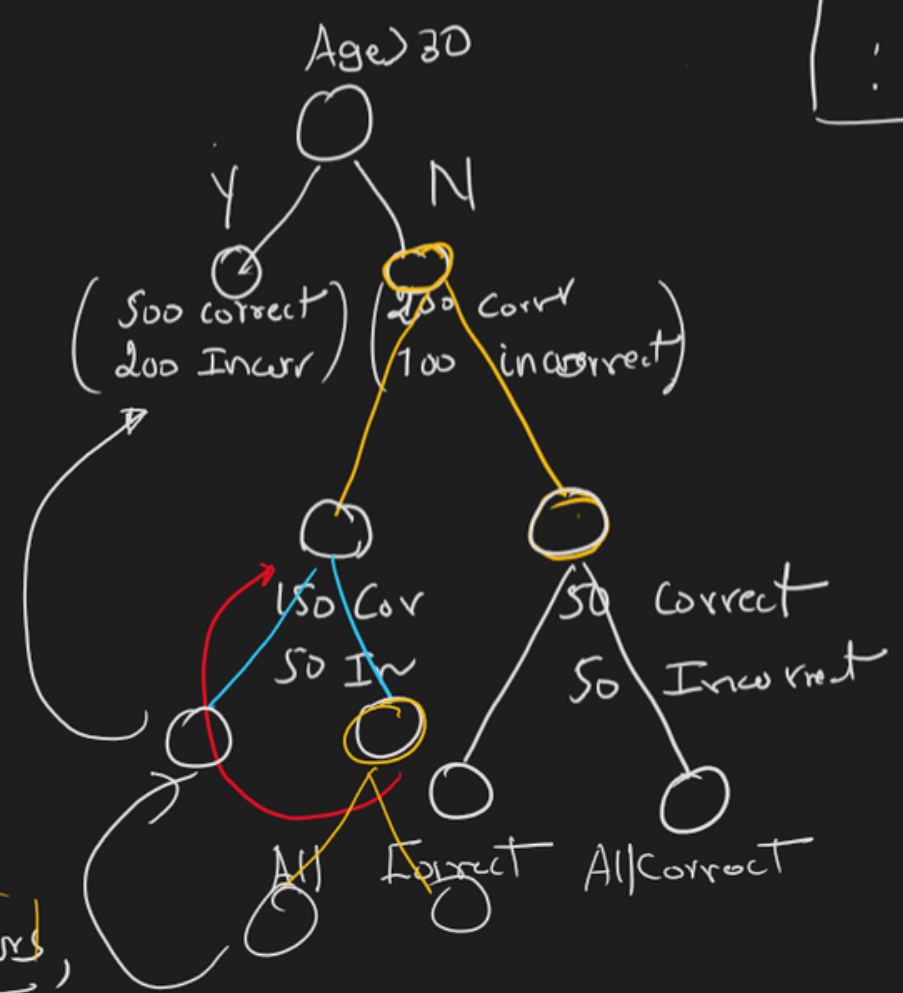
How to find the best split.

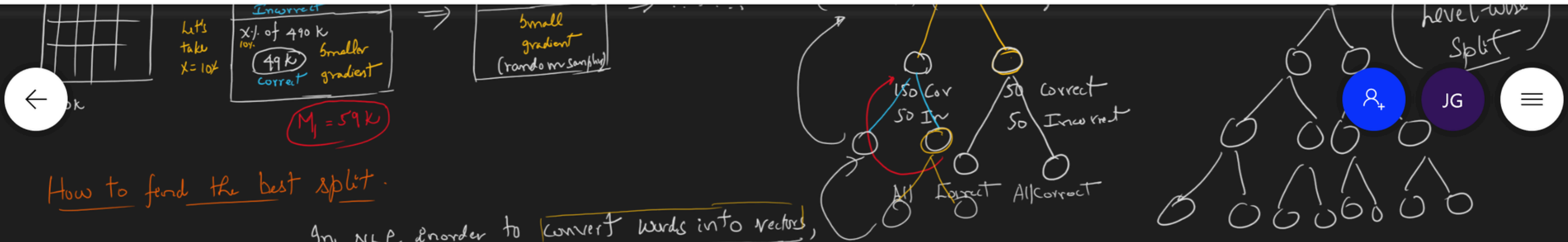
In NLP, in order to convert words into vectors, we use basic models like Bag of Models (BOM) &

Eligible/NE

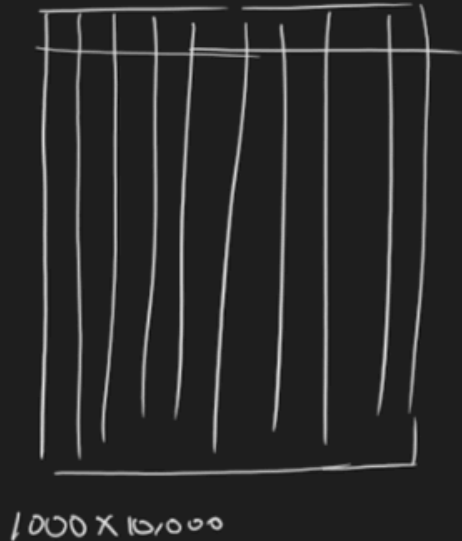
Age	Sal	Eligible/NE
30	50K	E
50	20K	NE
:	:	:
:	:	:

5000 recd





How to find the best split.



In NLP, in order to convert words into vectors, we use basic models like Bag of Words (BOW) & Term Frequency & inverse document frequency (tfidf)

Eg: "Chetana & Sanjeevi are colleagues back in 1990's."
"Sumra & Sanja are very close friends."

BOW

Sentiment Classification

	chetana	sanjeevi	are	colleagues	back	in	1990's	sumra	&	sanja	very	close	friends	y
sent1	1	1	1	1	1	1	1	0	1	0	0	0	0	1
sent2	0	0	0	0	0	0	0	1	1	1	1	1	1	1

2x13

{ Sparse Matrix / Sparse data => More '0's & less '1's }

sent2	0	0	0	0	0	0	0	1	1	1	1	1	1	1
-------	---	---	---	---	---	---	---	---	---	---	---	---	---	---

2x13

{ Sparse Matrix / Sparse data $\Rightarrow$ More '0's & less 1's }

$f_2 \leq 2$, Bin 1
 $2.0 \leq f_2 \leq 5$, Bin 2

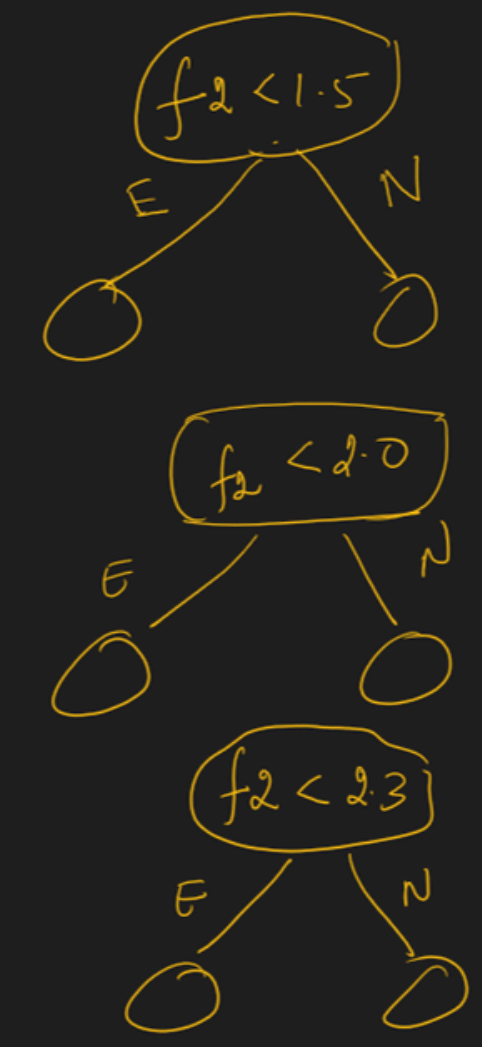
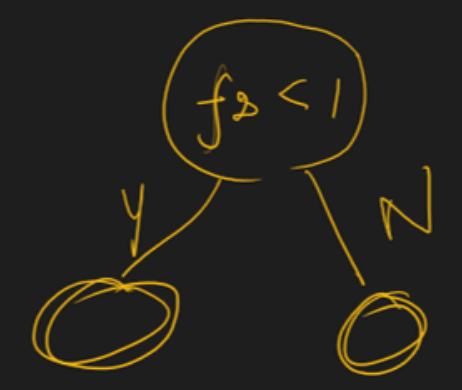
0
0.5
1.0
1.5
2.0
2.3
4.5

$\rightarrow 1$
 $\rightarrow 2$

f_2	y
1	E
1	N
2	E
1	N
1	E
1	E
2	N
2	E

f_1	f_2	f_3	f_4	y
-	1.5	-	-	E
-	2.0	-	-	N
-	2.3	-	-	E
-	1.0	-	-	N
-	0	-	-	E
-	0.5	-	-	E
-	2.3	-	-	N
-	4.5	-	-	E

1000



Best split? $\underline{IG} = ?$

Best split? $\underline{IG} = ?$

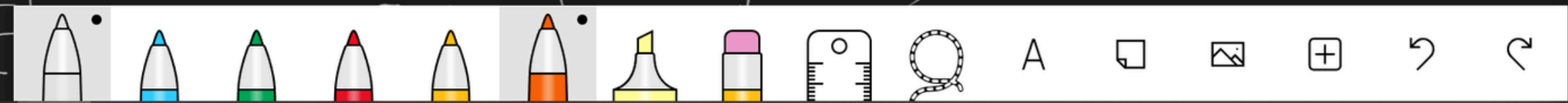
Histogram-based approach for splitting the tree

Drawbacks

① 5000 records $\rightarrow$ Bin 1 $\Rightarrow$ Accuracy $\downarrow \downarrow \downarrow$

of values

f_1	f_2	f_3



Drawbacks

①

f_1	f_2	f_3

10000 x 4

5000 records $\rightarrow$ Bin 1
5000 records $\rightarrow$ Bin 2

$\Rightarrow$ Accuracy $\downarrow \downarrow \downarrow$

② If I increase BIN size, DT needs to go for lot of calcu. $\Rightarrow$ Speed $\downarrow \downarrow \downarrow$

Trade-off b/w Speed & Accuracy

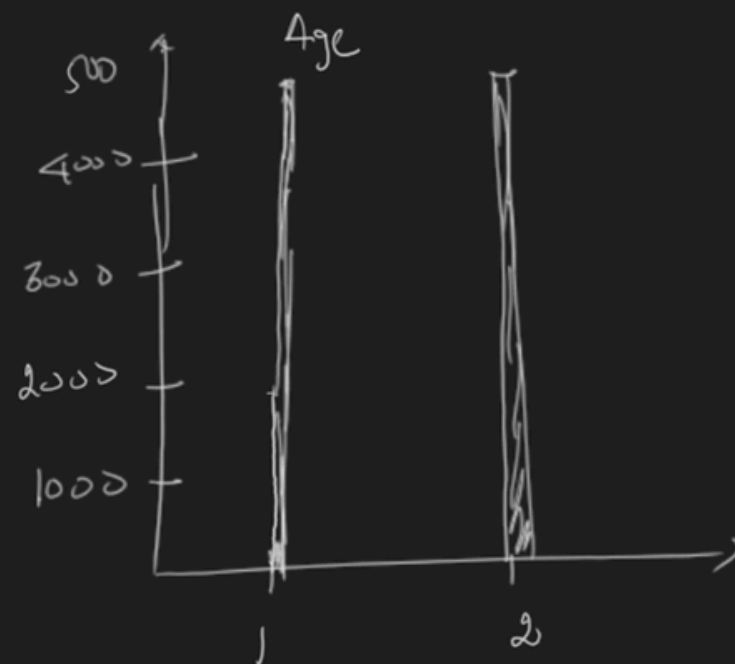
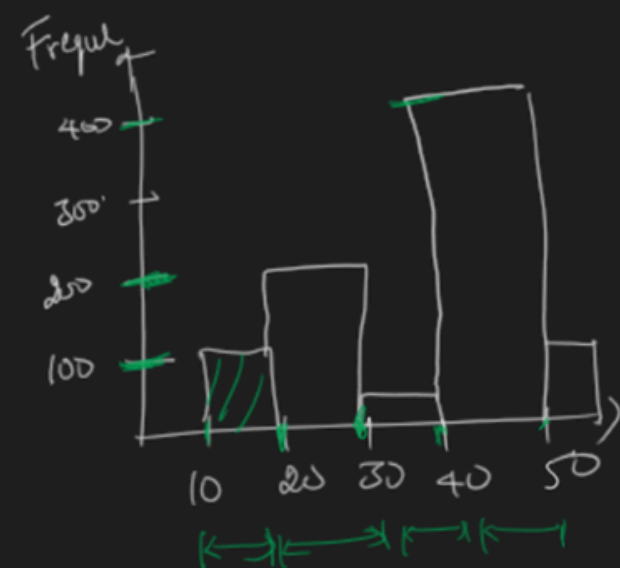
Feature Bundling $\Rightarrow$ "Identifying the possibility of combining one or more features"

f_1	f_2	f_3	...	f_n
1	0	10		
2	5	0	...	

$\Rightarrow$

f_1	$f_2 \& f_3$	f_3	...	f_n
1	10			
2	15			

bin $\Rightarrow$ range of values



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